

Lipid-derived Autacoids

Eicosanoids And Platelet activating factor

Learning outcomes

- **Demonstrate knowledge of pharmacological properties, mechanisms of action, and therapeutic uses of autacoids and anti-inflammatory drugs.**
- **Explain the mechanisms of action and therapeutic applications of autacoids and anti-inflammatory agents.**
- **Apply pharmacological knowledge to interpret clinical cases.**
- **Select appropriate drug for optimal therapy.**

Kinins

Kinins are formed from proteins called kininogens.

These are local inflammatory mediators—they cause:

Pain

Vasodilation

Increased vascular permeability

Kinins (Bradykinin, kallidin)

- **Vasodilator polypeptides**
- **Precursor: kininogen, α_2 globulin synthesized in the liver, circulate in plasma**
- **Plasma prekallikrein $\rightarrow$ plasma kallikrein, reaction catalyzed by hageman factor (factor XII)**

Kininogen $\rightarrow$ prekallikrein $\rightarrow$ kallikrein $\rightarrow$ bradykinin

Kinins: function

- Have role in inflammatory and hypersensitivity reactions
 - In mediating pain (nociception)
 - Regulation of blood pressure
 - Maintenance of fluid and electrolyte balance by the kidney
 - Spasmogenic (GIT & bronchi)
 - Vasodilatation, ↑ capillary permeability
- VD** of vascular bed (10 times > histamine)

Kinins (Bradykinin, kallidin)

HMW Kininogen

LMW kininogen

plasma Kallikrein

tissue kallikrein

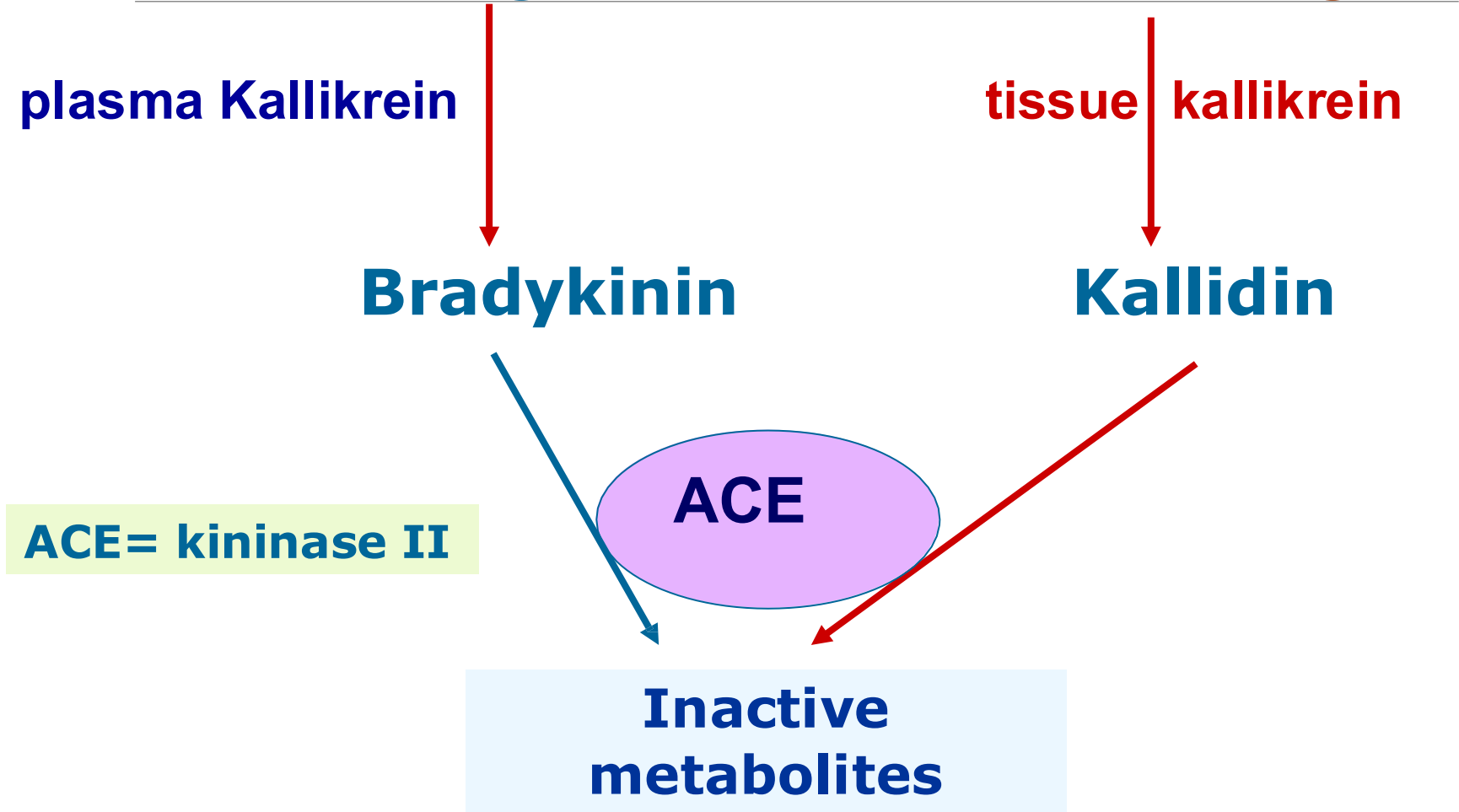
Bradykinin

Kallidin

ACE= kininase II

ACE

**Inactive
metabolites**



Drugs affecting:

1. NSAIDs :

NSAIDs do NOT change bradykinin (BK) levels directly

Bradykinin causes pain

But its pain effect is stronger when prostaglandins are present

👉 NSAIDs block prostaglandins

- Bradykinin is still there !
- But pain sensation is reduced as NSAIDs block prostaglandins

2. ACEIs

Because ACE normally breaks down bradykinin

If we block ACE → ↑ bradykinin → dry cough + angioedema

3. Aprotinin

- **Class:** Kallikrein inhibitor
- **Mechanism:** Inhibits kallikrein → ↓ bradykinin formation
- **Effect:** Reduces inflammation and bleeding
- **Use:** ↓ blood loss during major surgeries (e.g., cardiac surgery)
- **Key idea:** Blocks formation of bradykinin

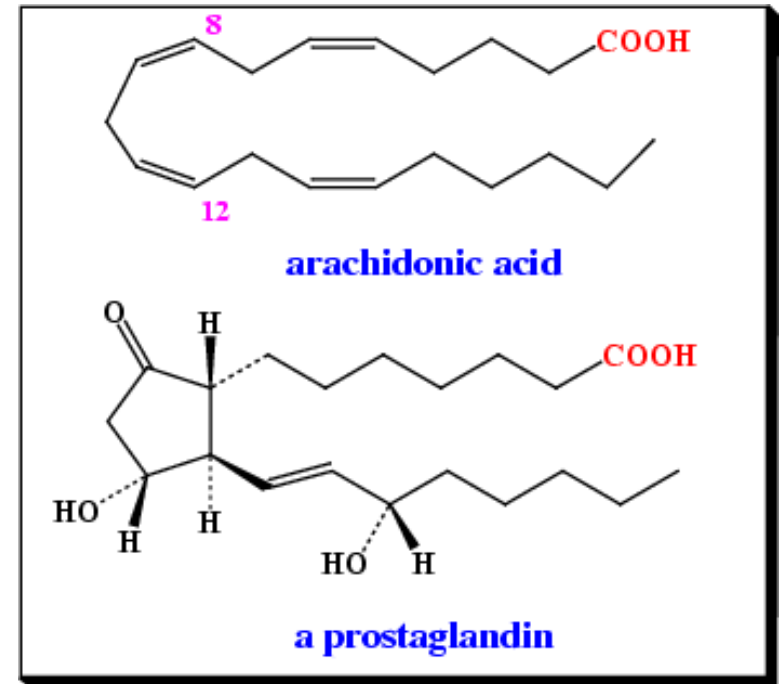
Eicosanoids

Eicosa=20 in Greek

a name given to a group of 20 carbon unsaturated fatty acids derived from arachidonic acid

they include:

Prostaglandins, thromboxanes, leukotriens



Eicosanoids

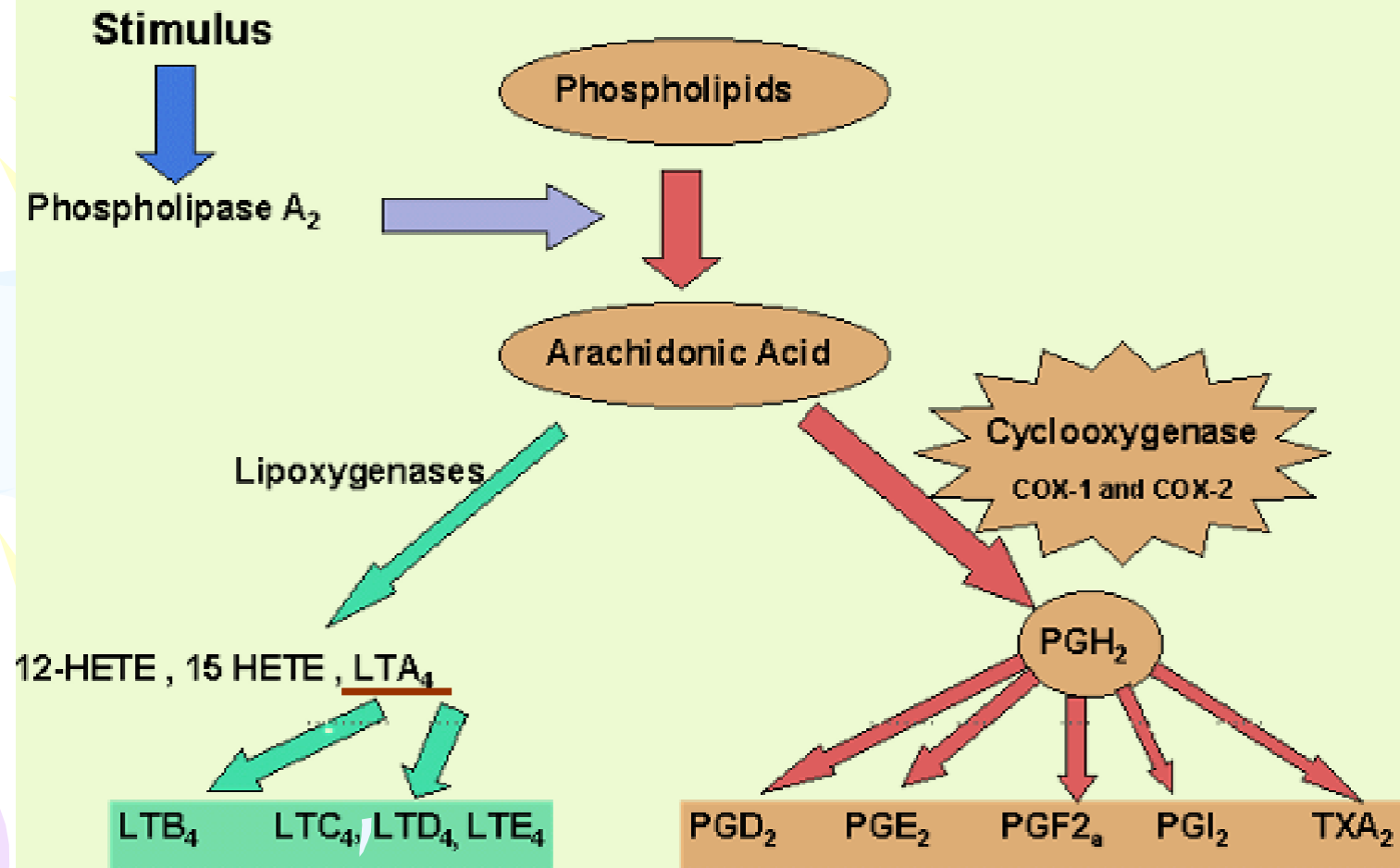
- Present in nearly all tissues
- Very high concentrations in **seminal fluid**
- Synthesized *de novo*
- Rapidly inactivated , **very short $t_{1/2}$**
- **PGE₂** inactivated in the lung (first pass metabolism)

Arachidonic acid metabolism

Two main oxidative pathways:

1. Cyclooxygenase pathway
prostaglandins, thromboxanes
2. 5-lipoxygenase pathway
leukotrienes

Figure 2 : Biosynthesis of eicosanoids



Physiological functions of prostaglandins

- Pain in tissue injury or inflammation
- Pyrogenic effect: fever associated with infections, tissue injury, inflammation,...
- ↑vascular permeability, produce local edema
- + TXA₂, LTs, PAF, histamine: mediators of bronchial asthma
- Control platelet aggregation
- Triggering agents for the onset of labor
- Prostaglandins in the stomach have a protective role
- Patency of ductus arteriosus (keep the **ductus arteriosus open**)
- It regulate renal blood flow and stimulate renin release

PGE

PGE mainly causes:

1. Vasodilation
2. Smooth muscle relaxation (but can contract uterus at low dose)
3. luteotropic (↑luteal progesterone secretion, causes uterine contraction at low doses)
4. Increases progesterone secretion

PGF

PGF mainly causes:

1. Vasoconstriction
2. Smooth muscle contraction (especially uterus)”

 Important:

Causes uterine contraction

 “PGF = contraction”

Used in:

- **Abortion**
- **Dysmenorrhea (painful menstruation)**

PGD

1. Bronchoconstriction

2. Relax renal BV

👉 “PGD = lung effect (bronchoconstriction)”

PGI₂ (Prostacyclin)

PGI₂ is produced by blood vessels

 Effects:

1. Vasodilation
2. ↓ platelet aggregation
3. Inhibits GIT secretions

 **“PGI₂ = anti-clot + vasodilation”**

TXA₂ (Thromboxane)

TXA₂ is produced by platelets

Effects:

1. Vasoconstriction
2. ↑ platelet aggregation

👉 TXA₂ = clot formation

Therapeutic uses

1) Abortion induction of labor:

Natural prostaglandins (**dinoprost, dinoprostone** (PGE_2)
& **carboprost** ($\text{PGF}_{2\alpha}$) --- causes abortion in 2nd trimester

Misoprostol (PGE_1) + **methotrexate**----- *terminate pregnancy
in 1st trimester.*

2) Peptic ulcer:

Misoprostol (synthetic PGE_1)-----↓ HCL & ↓ pepsin secretion
↑ mucosal resistance.

Used in patients chronically taking NSAIDs

Therapeutic uses

3) **Erectile dysfunction:**

Alprostadil (PGE₁)

- Vasodilation (V.D)
- Relaxation of smooth muscle

Therapeutic uses

- 4) To maintain patency of ductus arteriosus (PGE_1)
- 5) Glaucoma: Latanoprost ($\text{PgF}_{2\alpha}$), prodrug
- 6) primary pulmonary hypertension: PGI_2

Leukotriens

- Leukotrienes come from **LOX pathway**
- Produced mainly in **leukocytes**
- **LTB₄ = inflammation + neutrophil attraction**

Leukotriens: pharmacological actions

- Cysteinyl-containing leukotriens: (slow reacting substance of anaphylaxis): LTC₄, LTD₄, LTE₄
- Important in asthma

Effect:

- **Bronchoconstriction** (They cause strong narrowing of airways)
- **Increased permeability** (They make blood vessels leaky → swelling (edema))
- **Vasoconstriction**

Memb.phospholipids



corticosteroid

Phospholipase A₂

Arachidonic acid

Zileuton

5-Lipoxygenase

NSAIDs

COX-1

COX-2

Leukotrienes

LTB

LTC

LTD



Montelukast

PGI₂

PG

TXA₂

PGE

PGF

Platelet activating factor

- Phospholipid synthesized by platelets, leukocytes, and mast cells.
- Has a role in inflammation, asthma and reproduction (ovulation and implantation)

Pharmacological actions:

- Vasodilator, ↑platelet aggregation
- Stimulate leukocytes to aggregate and release LTs and lysosomal enzymes (inflammation)
- Tracheal, uterine and intestinal smooth muscles contraction